

Fundamentals of Electronics

Equations

$$V = \frac{W}{Q} \quad \mathbf{F} = k_e \frac{Q_1 Q_2}{d^2} \quad I = \frac{Q}{t}$$

$$\text{Life of a battery} = \frac{\text{amp-hour rating}}{\text{amps drawn}}$$

$$R = \rho \frac{l}{A} [1 + \alpha_{20} \Delta T]$$

$$V = IR \quad G = \frac{1}{R} \quad P = \frac{W}{t}$$

$$P = \frac{W}{t} = \frac{QV}{t} = IV = \frac{V^2}{R} = I^2 R$$

$$W_{\text{in}} = W_{\text{lost}} + W_{\text{stored}} + W_{\text{out}}$$

$$P_{\text{in}} = P_{\text{lost}} + P_{\text{stored}} + P_{\text{out}}$$

$$W_{\text{stored}} = \frac{1}{2} CV^2 = \frac{Q^2}{2C} \quad W_{\text{stored}} = \frac{1}{2} LI^2$$

$$\eta = \frac{W_{\text{out}}}{W_{\text{in}}} = \frac{P_{\text{out}}}{P_{\text{in}}}$$

$$\eta_{\text{total}} = \eta_1 \times \eta_2 \times \dots \times \eta_n$$

$$\text{Voltage Regulation} = \frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{full load}}}$$

$$\varepsilon = \frac{\mathbf{F}}{Q} = \frac{V}{d} \quad C = \frac{Q}{V} = \varepsilon \frac{A}{d} \quad \varepsilon_R = \frac{\varepsilon}{\varepsilon_0}$$

$$\tau_C = RC \quad \tau_L = \frac{L}{R} \quad L = \frac{MN^2 A}{l}$$

$$i_C = C \frac{d}{dt} v_C \quad v_{L, \text{avg}} = \frac{\Delta i_L}{\Delta t}$$

$$t = -\tau \ln \left(1 - \frac{v_C}{V_{\text{max}}} \right)$$

$$t = -\tau \ln \left(1 - \frac{i_L}{I_{\text{max}}} \right)$$

$$v_C = v_f - (v_f - v_i) e^{-t/\tau}$$

$$i_L = I_i - (I_f - I_i) e^{-t/\tau}$$

Charging:

$$v_C = E(1 - e^{-t/\tau}) \quad i_C = \frac{E}{R_T} e^{-t/\tau}$$

Discharging:

$$v_C = E e^{-t/\tau} \quad i_C = \frac{E}{R_T} (1 - e^{-t/\tau})$$

For Maximum power transfer:

$$R_L = R_{TH} \quad P_{L, \text{max}} = \frac{V_{TH}^2}{4R_{TH}}$$

Alternating Current

$$I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}} \quad V_{\text{rms}} = \frac{V_{\text{max}}}{\sqrt{2}}$$

$$\omega = 2\pi f \quad T = \frac{1}{f} \quad \mathbf{Y} = \frac{1}{\mathbf{Z}}$$

$$X_L = \omega L \quad X_C = \frac{1}{\omega C} \quad B = \frac{1}{X}$$

$$G = (\text{sum of areas}) / (\text{length of the curve})$$

$$pf = \cos|\theta_v - \theta_i| \quad P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} pf$$

pf leading: the current leads the voltage

pf lagging: the current lags the voltage

Impedance

The complex opposition to changing current/voltage.

$$\mathbf{Z}_C = R_C + jX_C \quad \mathbf{Z}_L = R_L + jX_L$$

Where:

- R is the real form (resistance) $[\Omega]$
- X is the imaginary form (reactance) $[\Omega]$

Admittance

The complex allowance of changing current/voltage.

$$\mathbf{Y}_C = G_C + jB_C \quad \mathbf{Y}_L = G_L + jB_L$$

Where:

- G is the real form (conductance) $[\text{S}]$
- B is the imaginary form (susceptance) $[\text{S}]$

Series Circuits

$$\sum V_{\text{closed loop}} = 0 \quad \frac{V_x}{V_S} = \frac{R_x}{R_T}$$

$$R_T = R_1 + R_2 + \dots + R_n$$

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

$$L_T = L_1 + L_2 + \dots + L_N$$

$$\mathbf{Z}_T = \mathbf{Z}_1 + \mathbf{Z}_2 + \dots + \mathbf{Z}_N$$

Parallel Circuits

$$\sum I_{\text{in}} = \sum I_{\text{out}} \quad I_S = I_1 + I_2 + \dots + I_n$$

$$I_S R_T = I_x R_x \quad R_T = \frac{R_1 R_2}{R_1 + R_2}$$

$$P_S = P_{R_1} + P_{R_2} + \dots + P_{R_n}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

$$C_T = C_1 + C_2 + \dots + C_N$$

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

$$\frac{1}{\mathbf{Z}_T} = \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \dots + \frac{1}{\mathbf{Z}_N}$$

Network Theorems

Superposition

1. Activate one source at a time
 - (a) Voltage source: short it
 - (b) Current source: open it
2. Find I'_1, I'_2 , etc.
3. Add I primes together to find current values

Mesh Current Analysis

1. Assume current directions
2. Write an equation per loop using KVL
3. Solve for unknown current(s)/value(s)

Super Mesh

Only if you have a current source in the middle

1. Assume current directions
2. Deactivate (open) the current source
3. Write one equation using both I_1 and I_2
4. For the other equation, use the sum of I_1 and I_2
5. Solve the system of equations

Nodal Analysis

1. Determine the number of nodes
2. Pick and label each node
3. Apply KCL at each node assuming each node has largest voltage value
4. Solve system of equations

Super Node

only if you have a voltage source preventing nodal analysis

1. Short the source
2. Proceed as though nodal analysis would work
3. Solve the system of equations

Thevenin's Theorem

1. Identify and remove the load
2. Find R_{TH} by deactivating all sources simultaneously
3. Find V_{TH} across terminals from step 1
4. Draw the equivalent circuit

Norton's Theorem

1. Identify and remove the load
2. Find R_N by deactivating all sources simultaneously
3. Find V_N by shorting terminals from step 1
4. Draw the equivalent Norton circuit